

Respostas/Soluções

1) a) $(101010)_2 + (111)_2 = (49)_{10}$

$$\begin{array}{r} 101010 \\ + 111 \\ \hline 110001 \end{array}$$

$$(190001)_2 \rightarrow ()_{10}$$

$$= 1 \times 2^5 + 1 \times 2^4 + 1$$

$$= 32 + 16 + 1$$

$$= (49)_{10}$$

b) $(FC71B)_{16} - (B2A6)_{16} = ()_2$

$$\begin{array}{r} FC71B \\ - B2A6 \\ \hline F1475 \end{array}$$

$$(11110001010001110101)_2$$

2)

 $A < B$

A_1	A_0	B_1	B_0	S
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$S = \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + \bar{A}_1 \bar{A}_0 B_1 \bar{B}_0 + \bar{A}_1 \bar{A}_0 B_1 B_0 + \bar{A}_1 A_0 B_1 \bar{B}_0 +$$

$$\bar{A}_1 A_0 B_1 B_0 + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 \bar{A}_0 (\bar{B}_1 B_0 + B_1 \bar{B}_0 + B_1 B_0) + \bar{A}_1 A_0 (B_1 \bar{B}_0 + B_1 B_0) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 \bar{A}_0 (\bar{B}_1 B_0 + B_1 (\bar{B}_0 + B_0)) + \bar{A}_1 A_0 (B_1 (\bar{B}_0 + B_0)) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 \bar{A}_0 (\bar{B}_1 B_0 + B_1) + \bar{A}_1 A_0 (B_1) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 \bar{A}_0 (B_1 + B_0) + \bar{A}_1 A_0 B_1 + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 \bar{A}_0 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_1 A_0 B_1 + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 (\bar{A}_0 B_1 + \bar{A}_0 B_0 + A_0 B_1) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 (B_1 (\bar{A}_0 + A_0) + \bar{A}_0 B_0) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 (B_1 + \bar{A}_0 B_0) + A_1 \bar{A}_0 B_1 B_0$$

$$S = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + A_1 \bar{A}_0 B_1 B_0$$

$$S = B_1 (\bar{A}_1 + A_1 \bar{A}_0 B_0) + \bar{A}_1 \bar{A}_0 B_0$$

$$S = B_1 (\bar{A}_1 + \bar{A}_0 B_0) + \bar{A}_1 \bar{A}_0 B_0$$

$$S = \bar{A}_1 B_1 + B_1 \bar{A}_0 B_0 + \bar{A}_1 \bar{A}_0 B_0$$

3) Tabela de Conversão (equivalência)

DEC	Excesso-3	BCD
	$E_3 E_2 E_1 E_0$	ABCD
0	0011	0000
1	0100	0001
2	0101	0010
3	0110	0011
4	0111	0100
5	1000	0101
6	1001	0110
7	1010	0111
8	1011	1000
9	1100	1001

TABELA VERDADE

$E_3 E_2 E_1 E_0$	A	B	C	D
0000	x	x	x	x
0001	x	x	x	x
0010	x	x	x	x
0011	0	0	0	0
0100	0	0	1	0
0101	0	0	1	1
0110	0	1	0	0
0111	0	1	0	1
1000	0	1	0	1
1001	0	1	1	0
1010	0	1	1	1
1011	1	0	0	0
1100	1	0	0	1
1101	x	x	x	x
1110	x	x	x	x
1111	x	x	x	x

A

	$\bar{E}_1$	E_1	
$\bar{E}_3$	x	x	0
E_3	0	0	0
	x	x	x
$\bar{E}_0$	0	0	0
E_0	0	0	0

$$A = E_3 E_2 + E_3 E_1 E_0$$

B

	$\bar{E}_1$	E_1	
$\bar{E}_3$	x	x	0
E_3	0	0	0
	x	x	x
$\bar{E}_0$	0	0	0
E_0	0	0	0

$$B = \bar{E}_2 E_1 E_0 + \bar{E}_2 \bar{E}_1 + \bar{E}_2 \bar{E}_0$$

C

	$\bar{E}_1$	E_1	
$\bar{E}_3$	x	0	0
E_3	0	0	1
	x	x	x
$\bar{E}_0$	0	0	0
E_0	0	0	0

$$C = \bar{E}_1 E_0 + E_1 \bar{E}_0 = E_1 \oplus E_0$$

D

	$\bar{E}_1$	E_1	
$\bar{E}_3$	x	x	0
E_3	1	0	0
	x	x	x
$\bar{E}_0$	1	0	0
E_0	1	0	0

$$D = \bar{E}_0$$

